

Sine, cosine, tangent and cotangent of angles from 0° to 180°

The ratios escape the right triangle, so that an obtuse angle can have a sine too.

LESSON 19–20

2 × 40 MIN

GEOMETRIYA 9, §14

EXTENSION BEYOND IGX

LESSON CLOCK

13'

23'

20'

20'

4'

Why the right triangle is not enough	13 min
The definition on a semicircle	23 min
Supplementary angles	20 min
The table	20 min
Homework	4 min

LEARNING OBJECTIVES

- Define the four ratios for any angle from 0° to 180° using the unit semicircle.
- Give the exact values at 0°, 30°, 45°, 60°, 90°, 120°, 135°, 150° and 180°.
- Use $\sin(180^\circ - \alpha) = \sin \alpha$ and $\cos(180^\circ - \alpha) = -\cos \alpha$.
- Determine the sign of each ratio for an obtuse angle.

TEXTBOOK

National	pp. 76–82
Cambridge	Core & Extended, pp. 300–310
Workbook	Exercise 15.1

01

Explanation

Why the right triangle is not enough

In Grade 8 $\sin A$ was defined as $\frac{\text{opposite}}{\text{hypotenuse}}$ in a right triangle, so A had to be acute. But a triangle can have an obtuse angle, and the sine rule of the next lesson will need its sine.

DEFINITION	WORKS FOR
ratios in a right triangle	$0^\circ < \alpha < 90^\circ$
coordinates on the unit semicircle	$0^\circ \leq \alpha \leq 180^\circ$
coordinates on the whole unit circle	every angle (Grade 9 algebra)

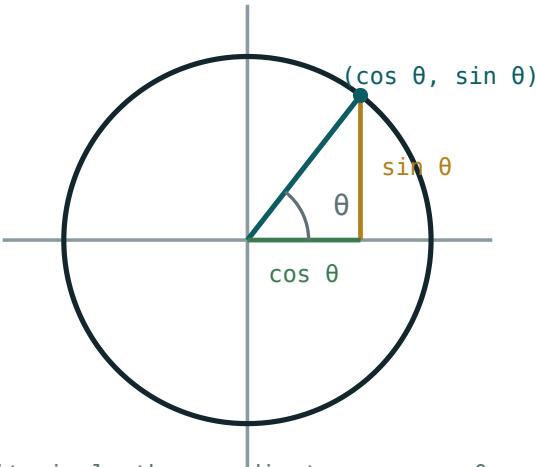
THE SAME EXTENSION AS IN ALGEBRA, ARRIVING FROM THE OTHER SIDE

The algebra course reaches the unit circle from rotation; geometry reaches it from the need to solve an obtuse triangle. The definition is identical, and the two courses meet here.

The definition on a semicircle

Draw the unit semicircle above Ox and let the ray at angle α from the positive x -axis meet it at $P(x, y)$. Then

$$\sin \alpha = y \quad \cos \alpha = x \quad \tan \alpha = \frac{y}{x} \quad \cot \alpha = \frac{x}{y}$$



on the unit circle the coordinates are $\cos \theta$ and $\sin \theta$

For an angle up to 180° the point lies on the upper half
— the ordinate is never negative.

RANGE	SIN	COS	TAN
$0^\circ < \alpha < 90^\circ$	+	+	+
$\alpha = 90^\circ$	1	0	undefined
$90^\circ < \alpha < 180^\circ$	+	−	−

SINE IS NEVER NEGATIVE FOR A TRIANGLE’S ANGLE

Every angle of a triangle lies strictly between 0° and 180° , so its point is on the upper semicircle and its sine is positive. That is exactly what makes the sine rule and the area formula work without any case analysis.

Supplementary angles

The points at α and $180^\circ - \alpha$ are mirror images in the y -axis, so the ordinates agree and the abscissae are opposite:

$$\sin(180^\circ - \alpha) = \sin \alpha \quad \cos(180^\circ - \alpha) = -\cos \alpha$$

$$\tan(180^\circ - \alpha) = -\tan \alpha \quad \cot(180^\circ - \alpha) = -\cot \alpha$$

ANGLE	REWRITE	SIN	COS
120°	$180^\circ - 60^\circ$	$\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$
135°	$180^\circ - 45^\circ$	$\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2}$
150°	$180^\circ - 30^\circ$	$\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$

TWO ANGLES SHARE A SINE; ONLY ONE SHARES A COSINE

$\sin 30^\circ = \sin 150^\circ$. This ambiguity is the reason the sine rule can give two answers for an angle, and why the cosine rule — whose cosine determines the angle uniquely — is preferred when there is any doubt.

The table

A	0°	30°	45°	60°	90°	120°	135°	150°	180°
$\sin$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\cos$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{3}}{2}$	-1
$\tan$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	—	$-\sqrt{3}$	-1	$-\frac{\sqrt{3}}{3}$	0

The sine row is symmetric about 90° ; the cosine row is antisymmetric. Reading the table that way halves what has to be remembered.

02 Worked examples**EXAMPLE 1** Find $\sin 120^\circ$, $\cos 120^\circ$ and $\tan 120^\circ$.

① $120^\circ = 180^\circ - 60^\circ$

② $\sin 120^\circ = \sin 60^\circ = \frac{\sqrt{3}}{2}$

Sine unchanged.

③ $\cos 120^\circ = -\cos 60^\circ = -\frac{1}{2}$

Cosine negated.

④ $\tan 120^\circ = -\sqrt{3}$

ANSWER

$$\frac{\sqrt{3}}{2}, -\frac{1}{2}, -\sqrt{3}$$

EXAMPLE 2 An angle of a triangle has $\sin \alpha = \frac{1}{2}$. What can α be?

① $\sin 30^\circ = \frac{1}{2}$

② $\sin 150^\circ = \sin(180^\circ - 30^\circ) = \frac{1}{2}$

③ Both lie strictly between 0° and 180° .

④ $\alpha = 30^\circ$ or $\alpha = 150^\circ$.
Two possibilities.

ANSWER

30° or 150°

EXAMPLE 3 An angle of a triangle has $\cos \alpha = -\frac{1}{2}$. Find α .

① A negative cosine means an obtuse angle.

② $\cos 60^\circ = \frac{1}{2}$

③ $\cos(180^\circ - 60^\circ) = -\frac{1}{2}$

④ $\alpha = 120^\circ$ — the only possibility.
Cosine is not ambiguous.

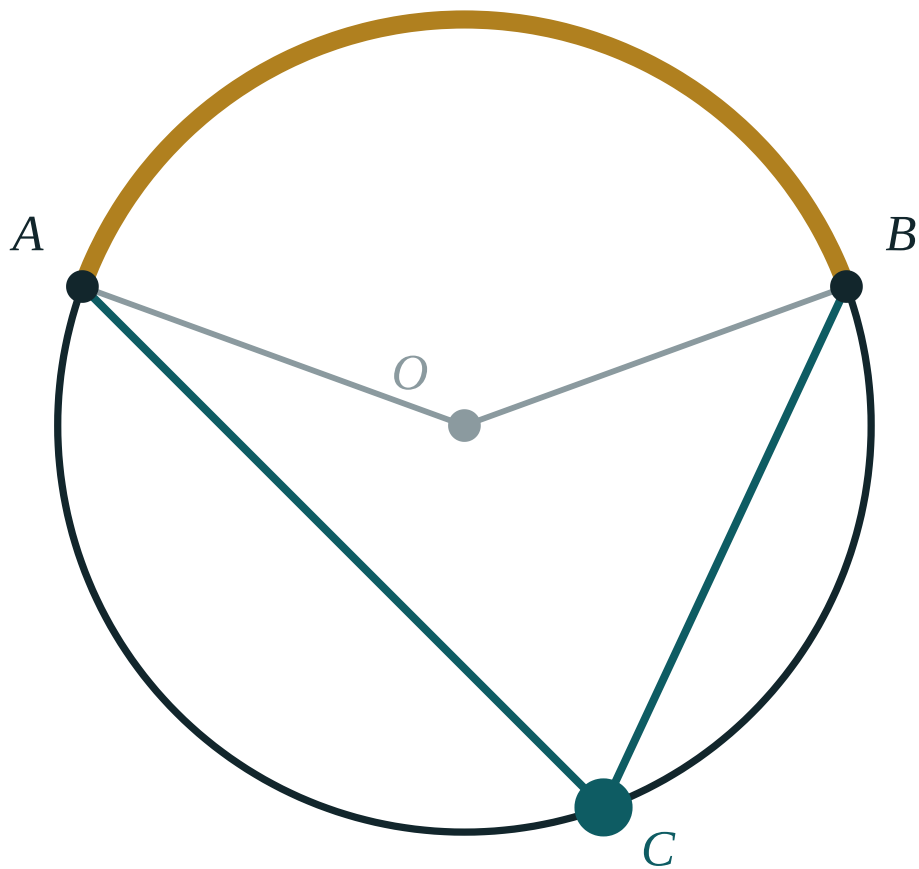
ANSWER

120°

03 Interactive model

Draw one semicircle and swing a ruler from 0° to 180° ; the class watches the height rise and fall while the horizontal reading passes through zero and turns negative.

From 0° to 180°



central $\angle AOB$ 140°

inscribed $\angle ACB$ 70°

ratio 2

arc AB 140°

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Obtuse angle	O'tmas burchak	Тупой угол
Unit semicircle	Birlik yarim aylana	Единичная полуокружность
Supplementary angles	Qo'shni burchaklar	Смежные углы
Ordinate	Ordinata	Ордината
Abcissa	Abssissa	Абсцисса
Sign	Ishora	Знак
Exact value	Aniq qiymat	Точное значение
Extension	Kengaytirish	Расширение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

For $0^\circ < \alpha < 180^\circ$, $\sin \alpha$ is:

always positive

always negative

sometimes zero

undefined

$\cos \alpha$ for an obtuse α is:

positive

negative

zero

undefined

$\sin(180^\circ - \alpha)$ equals:

$-\sin \alpha$

$\sin \alpha$

$\cos \alpha$

$-\cos \alpha$

$\cos(180^\circ - \alpha)$ equals:

$\cos \alpha$

$-\cos \alpha$

$\sin \alpha$

$-\sin \alpha$

$\cos 120^\circ$ equals:

$\frac{1}{2}$

$-\frac{1}{2}$

$\frac{\sqrt{3}}{2}$

$-\frac{\sqrt{3}}{2}$

$\sin \alpha = \frac{1}{2}$ in a triangle gives:

30° only

150° only

both

neither

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\sin 90^\circ$

ANSWER 1

2. $\cos 90^\circ$

ANSWER 0

3. $\sin 180^\circ$

ANSWER 0

4. $\cos 180^\circ$

ANSWER -1

5. $\sin 120^\circ$

ANSWER $\frac{\sqrt{3}}{2}$

6. $\cos 120^\circ$

ANSWER $-\frac{1}{2}$

7. $\sin 150^\circ$

ANSWER $\frac{1}{2}$

Medium · the standard question

7 PROBLEMS

1. $\cos 150^\circ$

ANSWER $-\frac{\sqrt{3}}{2}$

2. $\sin 135^\circ$

ANSWER $\frac{\sqrt{2}}{2}$

3. $\cos 135^\circ$

ANSWER $-\frac{\sqrt{2}}{2}$

4. $\tan 120^\circ$

ANSWER $-\sqrt{3}$

5. $\tan 135^\circ$

ANSWER -1

6. $\sin \alpha = \frac{1}{2}$ in a triangle

ANSWER 30° or 150°

7. $\cos \alpha = -\frac{1}{2}$ in a triangle

ANSWER 120°

Hard · stretch and reason

7 PROBLEMS

1. $\sin 120^\circ + \cos 60^\circ$

ANSWER $\frac{\sqrt{3} + 1}{2}$

2. $\sin 150^\circ \cdot \cos 150^\circ$

ANSWER $-\frac{\sqrt{3}}{4}$

3. $\sin^2 120^\circ + \cos^2 120^\circ$

ANSWER 1

4. $\cos \alpha = -0.8$ in a triangle: $\sin \alpha$

ANSWER 0.6

5. $\sin \alpha = 0.6$ and α obtuse: $\cos \alpha$

ANSWER -0.8

6. Which angle of a triangle can have a negative cosine?

ANSWER At most one — the obtuse angle

7. Simplify $\sin(180^\circ - \alpha) + \cos(180^\circ - \alpha)$

ANSWER $\sin \alpha - \cos \alpha$

07 Homework**Homework — 5 tasks**

Rewrite every obtuse angle as $180^\circ - \alpha$ before using the table.

1. Find $\sin 135^\circ$, $\cos 135^\circ$ and $\tan 135^\circ$.

2. Find $\sin 150^\circ$ and $\cos 150^\circ$.

3. An angle of a triangle has $\sin \alpha = \frac{\sqrt{2}}{2}$. What can α be?

4. An angle of a triangle has $\cos \alpha = -0.6$. Find $\sin \alpha$.

5. Explain why the sine of any angle of a triangle is positive.

The area of a triangle from two sides and the angle between them

$S = \frac{1}{2}ab \sin C$ — the formula that needs no height at all.

LESSON 21

1 × 40 MIN

GEOMETRIYA 9, §15

IGX 7.1

LESSON CLOCK

10'

14'

10'

6'

The derivation	10 min
Using it	14 min
Backwards	10 min
Homework	6 min

LEARNING OBJECTIVES

- Derive $S = \frac{1}{2}ab \sin C$ from the base-and-height formula.
- Use the formula for acute and obtuse included angles.
- Rearrange it to find a side or the angle.
- Apply it to parallelograms and to land measurement.

TEXTBOOK

National	pp. 83–86
Cambridge	Core & Extended, pp. 130–136
Workbook	Exercise 7.1

01

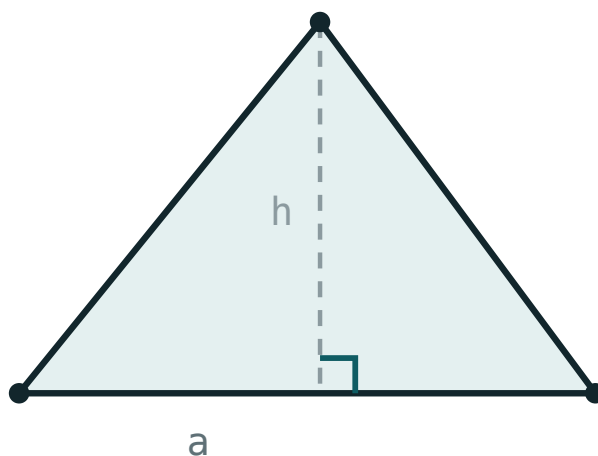
Explanation

The derivation

In $\triangle ABC$ drop the height h from A to BC . In the right triangle formed, $h = b \sin C$.

Substituting into $S = \frac{1}{2}ah$:

$$S = \frac{1}{2}ab \sin C$$



$$S = \frac{1}{2} \cdot a \cdot h$$

The height is $b \sin C$ — so the area needs only two sides and the angle between them.

The same argument works when C is obtuse: the foot of the height falls outside the triangle, but $h = b \sin(180^\circ - C) = b \sin C$ all the same. That is precisely why the last lesson was needed.

TWO SIDES AND THE ANGLE BETWEEN THEM

The angle must be the included one. $S = \frac{1}{2} ab \sin A$ is wrong unless A happens to lie between the sides a and b — which it does not, since a is opposite A .

Using it

SIDES	ANGLE	AREA
6, 8	30°	$\frac{1}{2} \cdot 48 \cdot 0.5 = 12$
6, 8	90°	24
6, 8	150°	12
5, 5	60°	$\frac{25\sqrt{3}}{4}$

The first and third rows have the same area, because $\sin 30^\circ = \sin 150^\circ$. The area is greatest when the angle is 90° — a fact worth remembering for optimisation questions.

For a **parallelogram** the same two sides and angle give twice as much: $S = ab \sin C$.

THE GREATEST AREA FOR TWO GIVEN SIDES IS AT 90°

Because $\sin C \leq 1$ with equality only at $C = 90^\circ$. Any “maximise the area” question about two fixed sides has the same one-word answer.

Backwards

Rearranging gives a side or the angle.

$$\sin C = \frac{2S}{ab} \quad b = \frac{2S}{a \sin C}$$

Example. $S = 12$, $a = 6$, $b = 8$. Then $\sin C = \frac{24}{48} = 0.5$, so $C = 30^\circ$ **or** $C = 150^\circ$.

GIVE BOTH ANSWERS UNLESS THE QUESTION RULES ONE OUT

A sine does not determine an angle of a triangle. Only extra information — “the triangle is acute”, or a diagram — removes one of the two. Losing the second answer is the standard mistake here and in the sine rule.

02 Worked examples

EXAMPLE 1 Find the area of a triangle with sides 6 and 8 and included angle 30° .

$$① \quad S = \frac{1}{2} \cdot 6 \cdot 8 \cdot \sin 30^\circ$$

$$② \quad = 24 \times 0.5$$

$$③ \quad = 12$$

$$④ \quad \text{The same as with } 150^\circ. \\ \sin 30^\circ = \sin 150^\circ.$$

ANSWER

12

EXAMPLE 2 A triangle has area 20 and sides 8 and 10 about the included angle. Find that angle.

$$① \quad \sin C = \frac{2 \times 20}{80}$$

$$② \quad = 0.5$$

$$③ \quad C = 30^\circ \text{ or } C = 150^\circ.$$

$$④ \quad \text{Both are possible without more information.}$$

ANSWER

30° or 150°

EXAMPLE 3 A parallelogram has sides 7 and 9 with an angle of 120° . Find its area.

$$① \quad S = ab \sin C \text{ for a parallelogram.}$$

Twice the triangle.

$$② \quad = 7 \times 9 \times \sin 120^\circ$$

$$③ \quad = 63 \times \frac{\sqrt{3}}{2}$$

$$④ \quad = \frac{63\sqrt{3}}{2} \approx 54.6$$

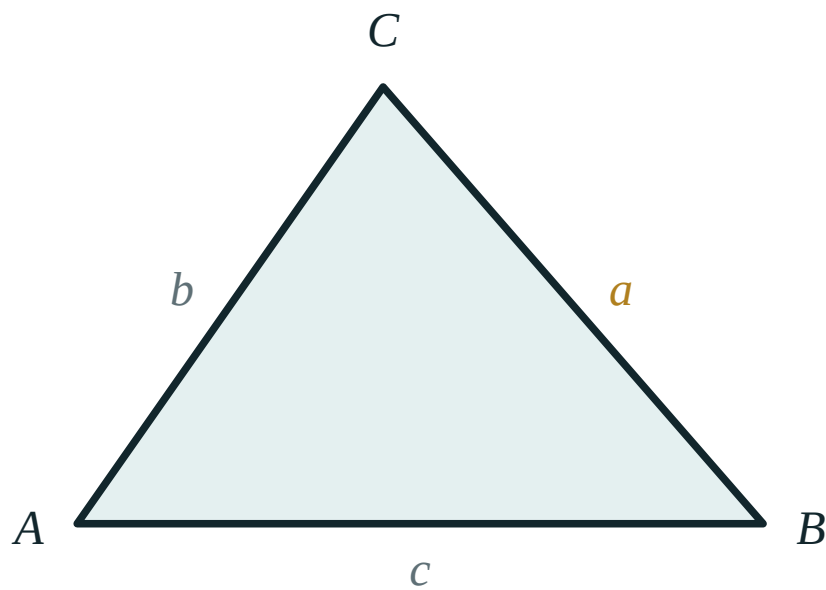
ANSWER

$$\frac{63\sqrt{3}}{2}$$

03 Interactive model

Fix two rods at a hinge and open them slowly; the area of the triangle they span grows to a maximum at a right angle and falls again — the formula made physical.

Two sides and the angle between



b 7

c 9

A 55°

a 7.6

B 49° C 76°

area 25.8

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Area of a triangle	Uchburchak yuzi	Площадь треугольника
Included angle	Orasidagi burchak	Угол между сторонами
Height	Balandlik	Высота
Base	Asos	Основание
To rearrange	Almashtirish	Выразить
Parallelogram	Parallelogramm	Параллелограмм
Square unit	Kvadrat birlik	Квадратная единица
Two answers	Ikki javob	Два ответа

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The area formula is:

$$\frac{1}{2}ab \sin C$$

$$ab \sin C$$

$$\frac{1}{2}ab \cos C$$

$$\frac{1}{2}a \sin C$$

The angle must be:

the largest

the included one

acute

opposite a

Sides 6, 8 and 30° :

$$12$$

$$24$$

$$48$$

$$6$$

The area is greatest when the angle is:

$\sin C = 0.5$ gives:

For a parallelogram the area is:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Sides 6, 8, angle 30°

ANSWER 12

2. Sides 6, 8, angle 90°

ANSWER 24

3. Sides 6, 8, angle 150°

ANSWER 12

4. Sides 10, 12, angle 30°

ANSWER 30

5. Sides 5, 5, angle 60°

ANSWER $\frac{25\sqrt{3}}{4}$

6. Parallelogram 4, 6, angle 30°

ANSWER 12

7. Greatest area for sides 6, 8

ANSWER 24

Medium · the standard question

7 PROBLEMS

1. Sides 7, 9, angle 120° : triangle area

ANSWER $\frac{63\sqrt{3}}{4}$

2. Same as a parallelogram

ANSWER $\frac{63\sqrt{3}}{2}$

3. $S = 20$, sides 8, 10: the angle

ANSWER 30° or 150°

4. $S = 24$, $a = 8$, $C = 30^\circ$: b

ANSWER 12

5. Sides 12, 5, angle 90°

ANSWER 30

6. An equilateral triangle of side a

ANSWER $\frac{a^2\sqrt{3}}{4}$

7. A rhombus of side 6 with a 60° angle

ANSWER $18\sqrt{3}$

Hard · stretch and reason

7 PROBLEMS

1. A field: two sides 120 m and 150 m with an angle of 75°

ANSWER $\approx 8693\text{ m}^2$

2. A triangle with $S = 30$, $a = 10$, $b = 12$: the angle

ANSWER 30° or 150°

3. A quadrilateral split by a diagonal into $\triangle(6, 8, 45^\circ)$ and $\triangle(8, 5, 60^\circ)$

ANSWER ≈ 34.3

4. A regular hexagon of side 4

ANSWER $24\sqrt{3}$

5. Two sides a and b : the angle giving half the maximum area

ANSWER 30° or 150°

6. A triangle of area 18 with sides 6 and 12

ANSWER 30° or 150°

7. Prove that the area of a parallelogram is $ab \sin C$

ANSWER Two congruent triangles

07 Homework

Homework — 5 tasks

When a sine gives two angles, write both unless the question excludes one.

- Find the area of a triangle with sides 9 and 14 and included angle 45° .
- A triangle has area 35 and sides 10 and 14. Find the included angle.
- A parallelogram has sides 5 and 8 with an angle of 150° . Find its area.
- Find the area of a regular hexagon of side 6.
- Show that the area of a triangle with two given sides is greatest when the included angle is 90° .

The sine rule

A side and its opposite angle move together — and the constant is the circumcircle’s diameter.

LESSON 22

1 × 40 MIN

GEOMETRIYA 9, §16

EXTENSION BEYOND IGX

LESSON CLOCK

10'

12'

12'

6'

The rule	10 min
When to use it	12 min
The ambiguous case	12 min
Homework	6 min

LEARNING OBJECTIVES

- State and prove the sine rule.
- Use it when a side and its opposite angle are known.
- Know that $a/\sin A = 2R$ and use it to find a circumradius.
- Recognise and resolve the ambiguous case.

TEXTBOOK

National	pp. 87–91
Cambridge	Core & Extended, pp. 310–316
Workbook	Exercise 15.2

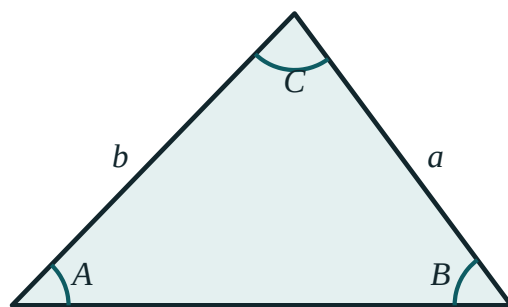
01

Explanation

The rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

where R is the radius of the circumscribed circle. The proof: the area of the triangle is $\frac{1}{2}ab\sin C = \frac{1}{2}bc\sin A = \frac{1}{2}ca\sin B$; dividing all three by $\frac{1}{2}abc$ gives the rule at once.



$$a / \sin A = b / \sin B = c / \sin C$$

$$a^2 = b^2 + c^2 - 2bc \cdot \cos A$$

Each side divided by the sine of its opposite angle gives the same number.

THE CONSTANT IS NOT ARBITRARY

That the common value is $2R$ is a genuinely useful fact: it turns any triangle question into a circle question. A triangle with $a = 6$ and $A = 30^\circ$ sits in a circle of radius 6, whatever its other two sides are.

When to use it

GIVEN	USE	FIND
two angles and any side (AAS, ASA)	the sine rule	the other sides
two sides and an angle opposite one of them (SSA)	the sine rule	the other angle — possibly two answers
two sides and the included angle (SAS)	the cosine rule	—
three sides (SSS)	the cosine rule	—

Example. $A = 40^\circ$, $B = 60^\circ$, $a = 8$. Then $C = 80^\circ$ and

$$b = \frac{a \sin B}{\sin A} = \frac{8 \sin 60^\circ}{\sin 40^\circ} \approx 10.8$$

PAIR A SIDE WITH ITS OWN OPPOSITE ANGLE

The rule links a with A , never a with B . If the given data contains no such matched pair, the sine rule cannot start and the cosine rule is the tool.

The ambiguous case

When two sides and a **non-included** angle are given, the sine found may correspond to two angles, and both may give a valid triangle.

Example. $a = 7, b = 9, A = 40^\circ$. Then

$$\sin B = \frac{9 \sin 40^\circ}{7} \approx 0.826$$

so $B \approx 55.7^\circ$ or $B \approx 124.3^\circ$. Both leave a positive third angle (84.3° or 15.7°), so **both triangles exist**.

CHECK	CONCLUSION
$\sin B > 1$	no triangle
the obtuse option makes the angle sum exceed 180°	one triangle
both options leave a positive third angle	two triangles
the side opposite the given angle is the longer	one triangle

ALWAYS TEST THE OBTUSE OPTION

Add it to the given angle: if the total is 180° or more, discard it; otherwise it is a real second answer. A question that expects two triangles and receives one loses half its marks.

02 Worked examples

EXAMPLE 1 In $\triangle ABC$, $A = 40^\circ$, $B = 60^\circ$, $a = 8$. Find b and C .

$$① \quad C = 180^\circ - 40^\circ - 60^\circ = 80^\circ$$

$$② \quad \frac{b}{\sin 60^\circ} = \frac{8}{\sin 40^\circ}$$

$$③ \quad b = \frac{8 \times 0.866}{0.643}$$

$$④ \quad b \approx 10.8$$

ANSWER

$$b \approx 10.8, C = 80^\circ$$

EXAMPLE 2 In $\triangle ABC$, $a = 7$, $b = 9$, $A = 40^\circ$. Find B .

$$① \quad \sin B = \frac{9 \sin 40^\circ}{7} \approx 0.826$$

$$② \quad B \approx 55.7^\circ \text{ or } B \approx 124.3^\circ.$$

$$③ \quad \text{Check: } 40 + 124.3 = 164.3 < 180 \quad \checkmark$$

Both survive.

$$④ \quad \text{Two triangles exist.}$$

ANSWER

$$B \approx 55.7^\circ \text{ or } 124.3^\circ$$

EXAMPLE 3 A triangle has $a = 10$ and $A = 30^\circ$. Find the radius of its circumscribed circle.

$$\textcircled{1} \quad \frac{a}{\sin A} = 2R$$

$$\textcircled{2} \quad 2R = \frac{10}{0.5}$$

$$\textcircled{3} \quad 2R = 20$$

$$\textcircled{4} \quad R = 10$$

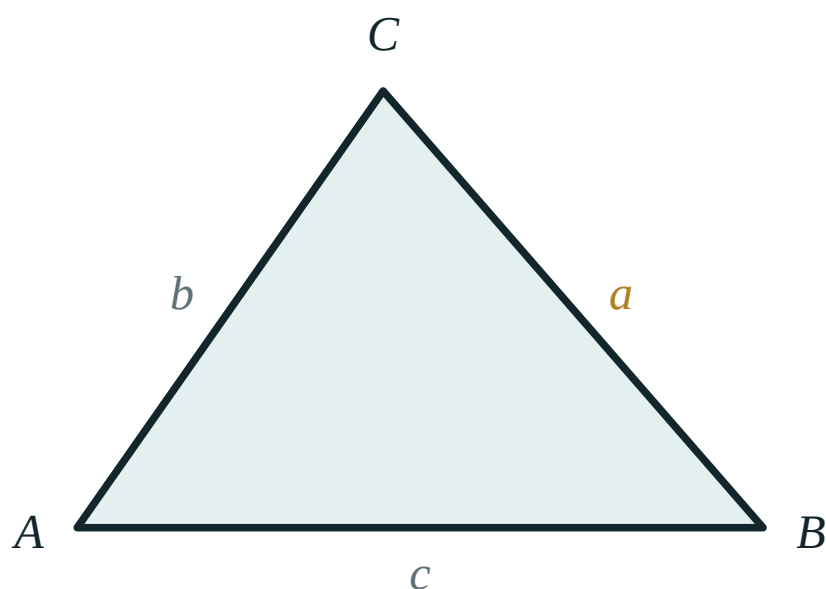
ANSWER

$$R = 10$$

03 Interactive model

Draw one circle and inscribe three different triangles on the same chord; the class sees that the angle opposite that chord never changes, which is the sine rule in a picture.

A side and its opposite angle



b 7

c 9

A 55°

a 7.6

B 49°

C 76°

area 25.8

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Sine rule	Sinuslar teoremasi	Теорема синусов
Opposite side	Qarshi tomon	Противолежащая сторона
Circumscribed circle	Tashqi chizilgan aylana	Описанная окружность
Circumradius	Tashqi radius	Радиус описанной окружности
Ambiguous case	Noaniq hol	Неоднозначный случай
Proportion	Proporsiya	Пропорция
Solve a triangle	Uchburchakni yechish	Решить треугольник
Given	Berilgan	Дано

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The sine rule links:

any side with any angle

a side with its opposite angle

two sides

two angles

The common value equals:

R

$2R$

R^2

πR

Use the sine rule when given:

SSS

SAS

AAS

nothing

The ambiguous case arises with:

$a = 10, A = 30^\circ: R =$

$\sin B > 1$ means:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. State the sine rule

ANSWER $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

2. The common value equals

ANSWER $2R$

3. $a = 10, A = 30^\circ: R$

ANSWER 10

4. $a = 8, A = 90^\circ: R$

ANSWER 4

5. $A = 40^\circ, B = 60^\circ: C$

ANSWER 80°

6. Use the sine rule for

ANSWER AAS or SSA

7. Use the cosine rule for

ANSWER SAS or SSS

Medium · the standard question

7 PROBLEMS

1. $A = 40^\circ, B = 60^\circ, a = 8: b$

ANSWER ≈ 10.8

2. Same: c

ANSWER ≈ 12.3

3. $a = 7, b = 9, A = 40^\circ: \sin B$

ANSWER ≈ 0.826

4. Same: B

ANSWER $\approx 55.7^\circ$ or 124.3°

5. $a = 12, A = 45^\circ, B = 60^\circ: b$

ANSWER ≈ 14.7

6. $a = 5, b = 8, A = 30^\circ: \sin B$

ANSWER 0.8

7. Same: how many triangles?

ANSWER Two

Hard · stretch and reason

7 PROBLEMS

1. $a = 9, b = 5, A = 60^\circ$: how many triangles?

ANSWER One — $a > b$

2. $a = 4, b = 9, A = 40^\circ$: how many triangles?

ANSWER None — $\sin B > 1$

3. A triangle inscribed in a circle of radius 6 has $A = 30^\circ$: a

ANSWER 6

4. An equilateral triangle of side a : its circumradius

ANSWER $\frac{a}{\sqrt{3}}$

5. $A = 45^\circ, B = 105^\circ, c = 10$: a

ANSWER ≈ 14.1

6. A triangle with $A = 30^\circ, B = 45^\circ$ and $a = 6$: its area

ANSWER ≈ 24.6

7. Prove $\frac{a}{\sin A} = 2R$

ANSWER The inscribed angle on the chord a

07 Homework

Homework — 5 tasks

Test the obtuse option in every SSA question, and say whether it survives.

- In $\triangle ABC$, $A = 50^\circ, B = 70^\circ, a = 12$. Find b and c .
- In $\triangle ABC$, $a = 6, b = 10, A = 30^\circ$. Find B , and say how many triangles exist.
- A triangle has $a = 14$ and $A = 45^\circ$. Find its circumradius.
- In $\triangle ABC$, $a = 5, b = 12, A = 60^\circ$. How many triangles are possible?

5. Prove the sine rule from the area formula.

The cosine rule

Pythagoras with a correction term — and the tool for every triangle the sine rule cannot start.

LESSON 23

1 × 40 MIN

GEOMETRIYA 9, §17

EXTENSION BEYOND IGC

LESSON CLOCK

10'	12'	12'	6'
The rule	10 min		
Two uses	12 min		
Classifying a triangle	12 min		
Homework	6 min		

LEARNING OBJECTIVES

- State and use the cosine rule in both of its forms.
- Use it for SAS and for SSS.
- See Pythagoras' theorem as the special case $C = 90^\circ$.
- Use the sign of $\cos C$ to classify a triangle as acute, right or obtuse.

TEXTBOOK

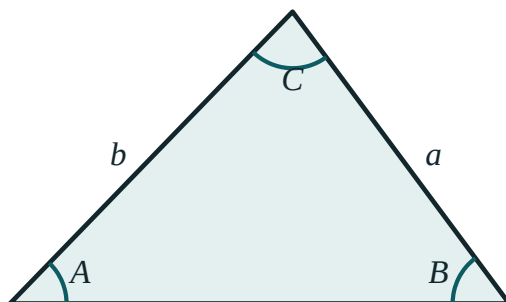
National	pp. 92–96
Cambridge	Core & Extended, pp. 317–322
Workbook	Exercise 15.3

01 Explanation

The rule

$$c^2 = a^2 + b^2 - 2ab \cos C$$

and likewise for the other two sides. When $C = 90^\circ$, $\cos C = 0$ and the rule becomes $c^2 = a^2 + b^2$ — Pythagoras' theorem is the special case.



$$a / \sin A = b / \sin B = c / \sin C$$

$$a^2 = b^2 + c^2 - 2bc \cdot \cos A$$

The cosine rule corrects Pythagoras by the term $-2ab \cos C$.

C	$\cos C$	EFFECT
$< 90^\circ$	> 0	$c^2 < a^2 + b^2$
$= 90^\circ$	0	$c^2 = a^2 + b^2$
$> 90^\circ$	< 0	$c^2 > a^2 + b^2$

UNLIKE THE SINE RULE, THE COSINE RULE IS NEVER AMBIGUOUS

A cosine determines an angle between 0° and 180° uniquely, sign and all. When both rules could be used to find an angle, the cosine rule is the safer choice.

Two uses

GIVEN	FORM TO USE	FINDS
SAS — two sides and the included angle	$c^2 = a^2 + b^2 - 2ab \cos C$	the third side
SSS — three sides	$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$	any angle

Example (SAS). $a = 7$, $b = 9$, $C = 60^\circ$:

$$c^2 = 49 + 81 - 2 \cdot 63 \cdot 0.5 = 130 - 63 = 67, \text{ so } c = \sqrt{67} \approx 8.19$$

Example (SSS). $a = 5$, $b = 6$, $c = 7$:

$$\cos C = \frac{25 + 36 - 49}{60} = \frac{12}{60} = 0.2, \text{ so } C \approx 78.5^\circ$$

THE ANGLE IN THE FORMULA IS OPPOSITE THE SIDE THAT STANDS ALONE

In $c^2 = a^2 + b^2 - 2ab \cos C$, the angle C and the side c are a matched pair. Using $\cos A$ with c on the left is the standard error.

Classifying a triangle

Compare the square of the longest side with the sum of the other two squares.

SIDES	TEST	TYPE
3, 4, 5	$25 = 9 + 16$	right
4, 5, 6	$36 < 16 + 25$	acute
4, 5, 8	$64 > 16 + 25$	obtuse

Only the longest side needs testing: the angle opposite it is the largest, and a triangle has at most one non-acute angle.

ONE INEQUALITY DECIDES THE WHOLE TRIANGLE

That is why the test is so quick, and it is a favourite short question in both the national and the Cambridge papers.

02 Worked examples

EXAMPLE 1 Find c when $a = 7$, $b = 9$, $C = 60^\circ$.

$$1 \quad c^2 = 7^2 + 9^2 - 2 \cdot 7 \cdot 9 \cdot \cos 60^\circ$$

$$2 \quad = 49 + 81 - 126 \times 0.5$$

$$3 \quad = 130 - 63 = 67$$

$$4 \quad c = \sqrt{67} \approx 8.19$$

ANSWER

$$c \approx 8.19$$

EXAMPLE 2 Find the largest angle of the triangle with sides 5, 6, 7.

① The largest angle is opposite 7.

② $\cos C = \frac{25 + 36 - 49}{2 \cdot 5 \cdot 6}$

③ $= \frac{12}{60} = 0.2$

④ $C \approx 78.5^\circ$
Acute — so the triangle is acute.

ANSWER $\approx 78.5^\circ$ **EXAMPLE 3** Classify the triangle with sides 4, 5, 8.

① Longest side 8: $8^2 = 64$.

② $4^2 + 5^2 = 16 + 25 = 41$

③ $64 > 41$

④ Obtuse.
The angle opposite 8 exceeds 90° .

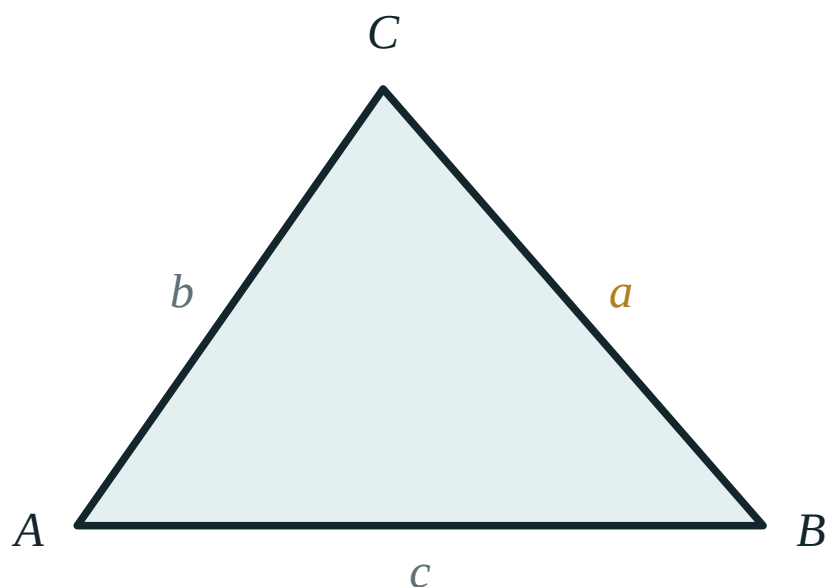
ANSWER

Obtuse

03 Interactive model

Fix two rods at 90° and read the third side; then open and close the angle and let the class see the correction term appear with each sign.

Pythagoras, corrected


b 7
c 9
A 55°
a 7.6
B 49°
C 76°
area 25.8

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Cosine rule	Kosinuslar teoremasi	Теорема косинусов
Correction term	Tuzatish hadi	Поправочный член
Special case	Xususiy hol	Частный случай
Acute triangle	O'tkir burchakli uchburchak	Остроугольный треугольник
Obtuse triangle	O'tmas burchakli uchburchak	Тупоугольный треугольник
Rearrange	Almashtirish	Выразить
Longest side	Eng katta tomon	Наибольшая сторона
Unique	Yagona	Единственный

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The cosine rule is:

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$c^2 = a^2 + b^2 + 2ab \cos C$$

$$c = a + b - 2ab \cos C$$

$$c^2 = a^2 - b^2$$

$C = 90^\circ$ gives:

the sine rule

Pythagoras

nothing

the area formula

Use it for:

AAS

SSA

SAS or SSS

AAA

Is the cosine rule ambiguous?

yes

no

sometimes

only for SSS

Sides 4, 5, 8 make a triangle that is:

acute

right

obtuse

impossible

$\cos C < 0$ means C is:

acute

right

obtuse

impossible

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. State the cosine rule

ANSWER $c^2 = a^2 + b^2 - 2ab \cos C$

2. $C = 90^\circ$ gives

ANSWER $c^2 = a^2 + b^2$

3. Use for SAS or

ANSWER SSS

4. Sides 3, 4, 5

ANSWER Right

5. Sides 4, 5, 6

ANSWER Acute

6. Sides 4, 5, 8

ANSWER Obtuse

7. $\cos C < 0$ means

ANSWER C is obtuse

Medium · the standard question

7 PROBLEMS

1. $a = 7, b = 9, C = 60^\circ: c$

ANSWER ≈ 8.19

2. $a = 5, b = 8, C = 120^\circ: c$

ANSWER ≈ 11.4

3. Sides 5, 6, 7: the largest angle

ANSWER $\approx 78.5^\circ$

4. Sides 3, 5, 7: the largest angle

ANSWER 120°

5. $a = 6, b = 10, C = 90^\circ: c$

ANSWER ≈ 11.66

6. Sides 2, 3, 4: $\cos C$ for the largest angle

ANSWER $-\frac{1}{4}$

7. Sides 7, 8, 9: type

ANSWER Acute

Hard · stretch and reason

7 PROBLEMS

1. Sides 3, 5, 7: the area

ANSWER $\frac{15\sqrt{3}}{4}$

2. A parallelogram with sides 5, 8 and an angle of 60° : the shorter diagonal

ANSWER 7

3. The longer diagonal

ANSWER $\sqrt{129}$

4. A triangle with $a = 8$, $b = 5$, $c = 7$: $\cos A$

ANSWER $\frac{1}{7}$

5. Two ships leave a port at 40° apart, at 12 and 18 km/h: their distance after 2 h

ANSWER ≈ 23.4 km

6. For which x is the triangle 5, 12, x right-angled?

ANSWER 13 or $\sqrt{119}$

7. A triangle with sides n , $n+1$, $n+2$ is right-angled for

ANSWER $n = 3$

07 Homework

Homework — 5 tasks

Pair the lone side with the angle in the formula; write the pair down before substituting.

- Find c when $a = 6$, $b = 10$, $C = 120^\circ$.
- Find the largest angle of the triangle with sides 4, 6, 8.
- Classify the triangles with sides 6, 8, 9 and 6, 8, 11.
- Two roads leave a junction at 55° . Find the distance between points 5 km and 8 km along them.

5. Show that the cosine rule gives Pythagoras' theorem when $C = 90^\circ$.

Applications of the sine and cosine rules

Bearings, heights, distances across a river — the rules put to work outside the exercise book.

LESSON 24–25

2 × 40 MIN

GEOMETRIYA 9, §18

EXTENSION BEYOND IGX

LESSON CLOCK

13'

23'

22'

18'

4'

Choosing the rule	13 min
Bearings	23 min
Heights and distances	22 min
Accuracy	18 min
Homework	4 min

LEARNING OBJECTIVES

- Choose between the two rules from the given data.
- Draw a labelled diagram from a worded description, including bearings.
- Solve problems on distance, height and navigation.
- Give answers to a sensible accuracy with units.

TEXTBOOK

National	pp. 97–103
Cambridge	Core & Extended, pp. 323–330
Workbook	Exercise 15.4

01

Explanation

Choosing the rule

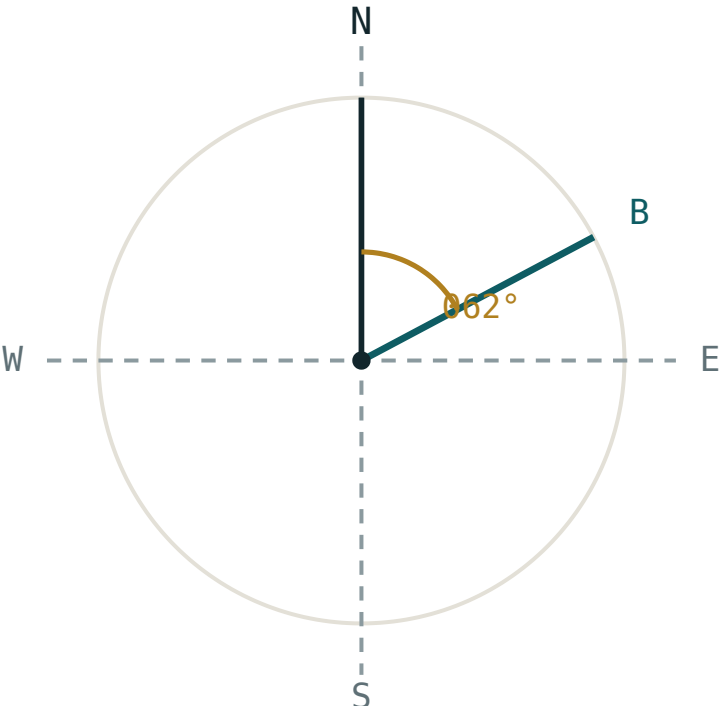
GIVEN	RULE	NOTE
two angles and a side	sine	find the third angle first
two sides and a non-included angle	sine	check for two answers
two sides and the included angle	cosine	gives the third side
three sides	cosine	gives any angle
two sides and the included angle, area wanted	$\frac{1}{2}ab \sin C$	no third side needed

LOOK FOR A MATCHED PAIR

A side together with its opposite angle means the sine rule can start. If the data contains no such pair, use the cosine rule — that decision takes five seconds and settles the whole question.

Bearings

A **bearing** is measured clockwise from due north and always written with three figures: 060° , 135° , 310° .



Bearings are measured clockwise from north, and written with three figures.

DIRECTION	BEARING
north	000°
east	090°
south	180°
west	270°

The **back bearing** — from B to A when the bearing from A to B is known — is found by adding 180° and subtracting 360° if the result exceeds it.

THE ANGLE IN THE TRIANGLE IS NOT THE BEARING

Two bearings of 060° and 140° from the same point give an angle of 80° between the paths. Always convert bearings into an angle of the triangle before applying a rule.

Heights and distances

Across a river. To find AB across water: mark a baseline AC on one bank, measure AC , and measure the angles at A and C to the far point B . Then the sine rule gives

AB.

A tower. From two points 50 m apart in line with the base, the angles of elevation are 30° and 45°. The triangle formed by the two observation points and the top has angles 30°, 135° and 15° — and the sine rule gives the slant distance, then the height.

STEP	WORKING
angle at the top	$45^\circ - 30^\circ = 15^\circ$
sine rule	$\frac{d}{\sin 30^\circ} = \frac{50}{\sin 15^\circ}$
d	≈ 96.6
height	$96.6 \sin 45^\circ \approx 68.3$

THE ANGLE AT THE TOP IS THE DIFFERENCE OF THE ELEVATIONS

It is the exterior-angle theorem in disguise, and it turns every two-observation tower problem into a single application of the sine rule.

Accuracy

DATA GIVEN TO	ANSWER TO
whole numbers	3 significant figures
1 decimal place	3 significant figures
exact values	exact, if possible

DO NOT ROUND IN THE MIDDLE

Rounding $\sin 15^\circ$ to 0.26 before dividing shifts the tower's height by more than a metre. Keep full accuracy in the calculator and round only the final answer — and always give the unit.

02 Worked examples

EXAMPLE 1 From two points 50 m apart in line with a tower, the angles of elevation are 30° and 45° . Find the height.

① Angle at the top = $45^\circ - 30^\circ = 15^\circ$.
Exterior angle.

②
$$\frac{d}{\sin 30^\circ} = \frac{50}{\sin 15^\circ}$$

③
$$d = \frac{50 \times 0.5}{0.2588} \approx 96.6 \text{ m}$$

④ Height = $96.6 \sin 45^\circ \approx 68.3 \text{ m}$.

ANSWER

$\approx 68.3 \text{ m}$

EXAMPLE 2 A ship sails 30 km on a bearing of 060° , then 40 km on 150° . Find its distance from the start.

① The turn is $150^\circ - 060^\circ = 90^\circ$.
The interior angle is 90° .

② $d^2 = 30^2 + 40^2$
The cosine rule with $\cos 90^\circ = 0$.

③ $= 900 + 1600 = 2500$

④ $d = 50 \text{ km}$

ANSWER

50 km

EXAMPLE

3

To find the width AB of a river, a baseline $AC = 60\text{ m}$ is measured with $\angle BAC = 70^\circ$ and $\angle BCA = 55^\circ$.

$$\textcircled{1} \quad \angle ABC = 180^\circ - 70^\circ - 55^\circ = 55^\circ.$$

$$\textcircled{2} \quad \frac{AB}{\sin 55^\circ} = \frac{60}{\sin 55^\circ}$$

$\textcircled{3}$ The two angles are equal, so $AB = AC$.
An isosceles triangle.

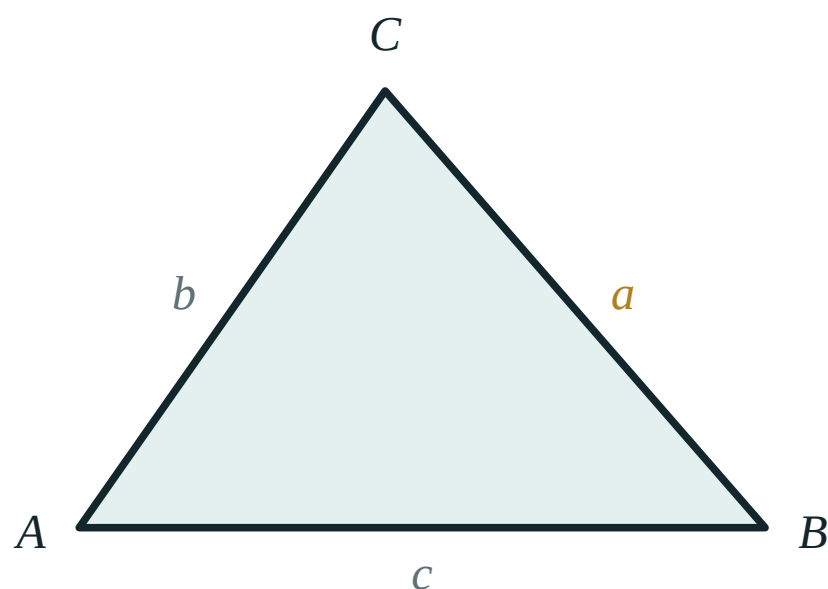
$$\textcircled{4} \quad AB = 60\text{ m}$$

ANSWER 60 m

03 Interactive model

Take the class outside with a protractor and a tape; measuring the school building by two elevations makes the sine rule an instrument in twenty minutes.

Two observations, one height



b 7

c 9

A 55°

a 7.6

B 49°

C 76°

area 25.8

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Bearing	Azimut	Азимут
Due north	Shimolga	На север
Angle of elevation	Ko'tarilish burchagi	Угол возвышения
Angle of depression	Tushish burchagi	Угол понижения
Navigation	Navigatsiya	Навигация
Accuracy	Aniqlik	Точность
Three-figure bearing	Uch xonali azimuth	Трёхзначный азимут
Sketch	Chizma	Чертёж

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A bearing is measured from:

east, anticlockwise

north, clockwise

north, anticlockwise

the horizontal

East is a bearing of:

000°

090°

180°

270°

Bearings 060° and 140° give an angle of:

60°

80°

140°

200°

Elevations 30° and 45° give a top angle of:

15°

75°

105°

135°

Two sides and the included angle: use

the sine rule

the cosine rule

Pythagoras

a bearing

Rounding should be done:

at every step

only at the end

never

to 1 s.f.

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Bearing of due east

ANSWER 090°

2. Bearing of due south

ANSWER 180°

3. Back bearing of 070°

ANSWER 250°

4. Back bearing of 200°

ANSWER 020°

5. Angle between bearings 060° and 140°

ANSWER 80°

6. Elevations 30° and 45° : the top angle

ANSWER 15°

7. Two sides and the included angle: rule

ANSWER Cosine

Medium · the standard question

7 PROBLEMS

1. Ship: 30 km on 060° , then 40 km on 150°

ANSWER 50 km

2. Tower: 50 m apart, elevations 30° and 45°

ANSWER ≈ 68.3 m

3. River: $AC = 60$, angles 70° and 55° : AB

ANSWER 60 m

4. Two roads at 55° , points 5 and 8 km along

ANSWER ≈ 6.6 km

5. A triangle plot 40, 55, 70 m: its largest angle

ANSWER $\approx 93.6^\circ$

6. Its area

ANSWER ≈ 1098 m²

7. A ladder 6 m at 70° to the ground: the height reached

ANSWER ≈ 5.64 m

Hard · stretch and reason

7 PROBLEMS

1. A ship sails 20 km on 040° then 35 km on 110° : its distance from the start

ANSWER $\approx 45.9\text{ km}$

2. Its bearing from the start

ANSWER $\approx 086^\circ$

3. A tower: elevations 25° and 40° from points 80 m apart

ANSWER $\approx 83.9\text{ m}$

4. Two lighthouses 12 km apart; a boat bears 040° from one and 310° from the other

ANSWER Solve with the sine rule

5. A triangle field $120, 150\text{ m}$ with an angle of 75° : the third side

ANSWER $\approx 166\text{ m}$

6. Its area

ANSWER $\approx 8693\text{ m}^2$

7. An aircraft flies 300 km on 070° then turns onto 160° for 400 km

ANSWER 500 km from the start

07 Homework

Homework — 5 tasks

Draw and label a diagram before any calculation; mark north on every bearing question.

1. A ship sails 25 km on 050° , then 40 km on 140° . Find its distance from the start.
2. From two points 60 m apart the angles of elevation of a tower are 35° and 50° . Find its height.
3. A triangular plot has sides $45, 60$ and 75 m . Find its largest angle and its area.
4. Two roads leave a junction at 65° . Find the distance between points 6 and 9 km along them.
5. Find the back bearing of 115° and of 295° .

The angle between two vectors and the scalar product

A multiplication of vectors whose answer is a number — and it measures the angle between them.

LESSON 26–27

2 × 40 MIN

GEOMETRIYA 9, §19

EXTENSION BEYOND IGX

LESSON CLOCK

13'

23'

22'

18'

4'

The angle between vectors	13 min
The scalar product	23 min
Finding an angle	22 min
Perpendicularity	18 min
Homework	4 min

LEARNING OBJECTIVES

- Define the angle between two vectors and the scalar product.
- Compute $\mathbf{a} \cdot \mathbf{b}$ from coordinates and from $|\mathbf{a}||\mathbf{b}| \cos \varphi$.
- Use the scalar product to find an angle and to test perpendicularity.
- Apply it to prove that two lines are perpendicular.

TEXTBOOK

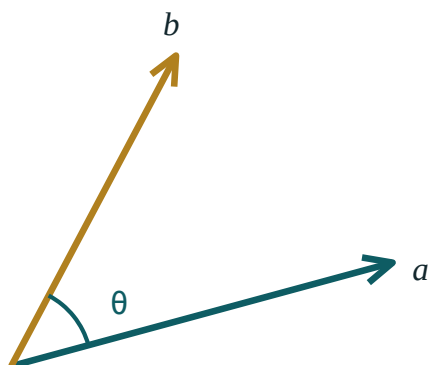
National	pp. 104–110
Cambridge	Core & Extended, pp. 331–338
Workbook	Exercise 15.5

01

Explanation

The angle between vectors

To measure the angle between $\vec{a}$ and $\vec{b}$, move them so that they start at the same point. The angle φ between them satisfies $0^\circ \leq \varphi \leq 180^\circ$ — which is exactly the range the first lesson of this chapter prepared.



$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta = a_1 b_1 + a_2 b_2 + a_3 b_3$$

perpendicular exactly when $\mathbf{a} \cdot \mathbf{b} = 0$

Brought to a common origin, two vectors make one angle between 0° and 180° .

Φ	THE VECTORS ARE
0°	parallel, same direction
90°	perpendicular
180°	parallel, opposite directions

A vector $\vec{a}(x, y)$ has magnitude $|\vec{a}| = \sqrt{x^2 + y^2}$ — the distance formula, applied to the arrow.

The scalar product

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \varphi$$

and, in coordinates,

$$\vec{a}(x_1, y_1) \cdot \vec{b}(x_2, y_2) = x_1 x_2 + y_1 y_2$$

The result is a **number**, not a vector — which is why it is called the *scalar* product.

PROPERTY	STATEMENT
commutative	$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$
distributive	$\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$
with a scalar	$(k\vec{a}) \cdot \vec{b} = k(\vec{a} \cdot \vec{b})$
square	$\vec{a} \cdot \vec{a} = \vec{a} ^2$

TWO FORMULAE FOR ONE QUANTITY – AND THAT IS THE POINT

One formula computes it from coordinates; the other contains the angle. Setting them equal is how an angle is found, and it is the only method available without a diagram.

Finding an angle

$$\cos \varphi = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$$

Example. $\vec{a}(3, 4)$ and $\vec{b}(1, 2)$. Then $\vec{a} \cdot \vec{b} = 3 + 8 = 11$, $|\vec{a}| = 5$, $|\vec{b}| = \sqrt{5}$, so

$$\cos \varphi = \frac{11}{5\sqrt{5}} \approx 0.9839, \varphi \approx 10.3^\circ$$

$\vec{A} \cdot \vec{B}$	Φ
> 0	acute
$= 0$	90°
< 0	obtuse

THE SIGN OF THE SCALAR PRODUCT IS THE SIGN OF THE COSINE

Because the two magnitudes are never negative. So a single subtraction tells you whether the angle is acute or obtuse, without computing it — a useful check on any answer.

Perpendicularity

$$\vec{a} \perp \vec{b} \iff \vec{a} \cdot \vec{b} = 0 \iff x_1x_2 + y_1y_2 = 0$$

This is the standard way to prove a right angle without measuring anything.

Example. Show that the triangle $A(1, 1)$, $B(4, 2)$, $C(2, 8)$ is right-angled.

VECTOR	COORDINATES
AB	$(3, 1)$
AC	$(1, 7)$
BC	$(-2, 6)$

$AB \cdot BC = -6 + 6 = 0$, so the right angle is at B .

ONE MULTIPLICATION AND ONE ADDITION

Compared with computing three side lengths and testing Pythagoras, the scalar product is far quicker — and it also tells you **where** the right angle is, which Pythagoras only reveals after the fact.

02 Worked examples

EXAMPLE 1 Find the angle between $\vec{a}(3, 4)$ and $\vec{b}(1, 2)$.

$$① \quad \vec{a} \cdot \vec{b} = 3 \times 1 + 4 \times 2 = 11$$

$$② \quad |\vec{a}| = 5, |\vec{b}| = \sqrt{5}$$

$$③ \quad \cos \varphi = \frac{11}{5\sqrt{5}} \approx 0.9839$$

$$④ \quad \varphi \approx 10.3^\circ$$

Acute, as the positive product predicted.

ANSWER

$$\approx 10.3^\circ$$

EXAMPLE 2 Show that $A(1, 1)$, $B(4, 2)$, $C(2, 8)$ is right-angled, and say where.

- ① $AB(3, 1)$, $BC(-2, 6)$.
From B in both directions.

② $AB \cdot BC = 3(-2) + 1(6)$

③ $= -6 + 6 = 0$

- ④ Right-angled at B .

ANSWER

Right-angled at B

EXAMPLE 3 For which t are $\vec{a}(t, 3)$ and $\vec{b}(4, -2)$ perpendicular?

① $\vec{a} \cdot \vec{b} = 4t - 6$

- ② Perpendicular means the product is zero.

③ $4t - 6 = 0$

④ $t = 1.5$

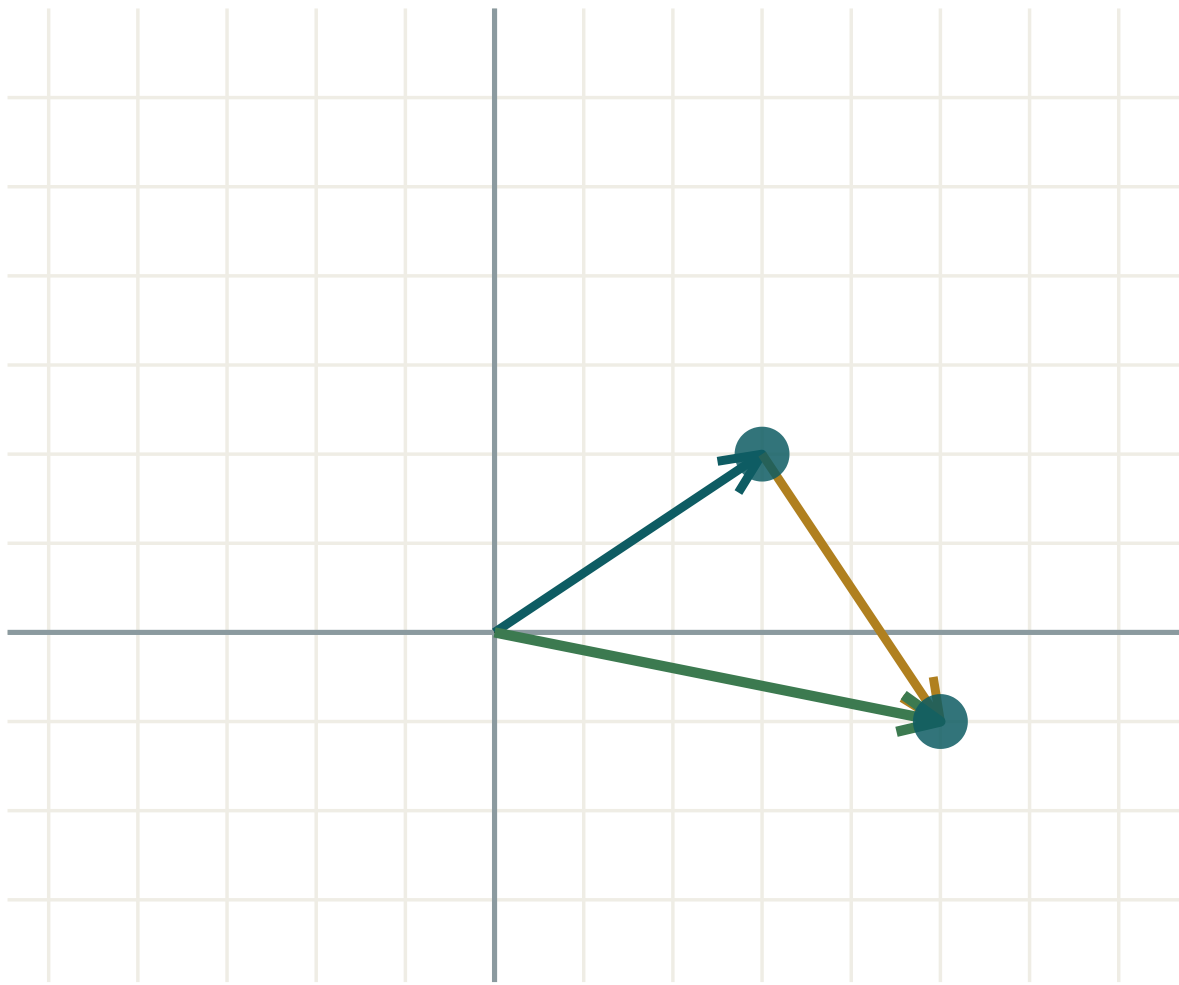
ANSWER

$t = 1.5$

03 Interactive model

Hold two metre rules at a common point and ask the class to predict the sign of the product as the angle opens past a right angle; the sign change is visible before any arithmetic.

Two arrows and one number



$a \ (3; \ 2)$

$b \ (2; \ -3)$

$a + b \ (5; \ -1)$

$|a| \ 3.61$

$|b| \ 3.61$

$|a + b| \ 5.1$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Vector	Vektor	Вектор
Scalar product	Skalyar ko'paytma	Скалярное произведение
Angle between vectors	Vektorlar orasidagi burchak	Угол между векторами
Magnitude	Modul	Модуль
Perpendicular	Perpendikulyar	Перпендикулярный
Coordinates of a vector	Vektor koordinatalari	Координаты вектора
Collinear	Kollinear	Коллинеарные
Unit vector	Birlik vektor	Единичный вектор

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The scalar product of two vectors is:

a vector

a number

an angle

a length

$\vec{a} \cdot \vec{b}$ in coordinates:

$x_1y_2 - x_2y_1$

$x_1x_2 + y_1y_2$

$x_1x_2 - y_1y_2$

$x_1 + y_2$

$\vec{a} \cdot \vec{b} = 0$ means:

parallel

perpendicular

equal

zero vectors

$\vec{a} \cdot \vec{b} < 0$ means φ is:

$\vec{a} \cdot \vec{a}$ equals:

$(3,4)$ has magnitude:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $(3,4) \cdot (1,2)$

ANSWER 11

2. $(2,3) \cdot (4,-1)$

ANSWER 5

3. $(1,0) \cdot (0,1)$

ANSWER 0

4. $|(3,4)|$

ANSWER 5

5. $|(5,12)|$

ANSWER 13

6. $\vec{a} \cdot \vec{a}$ for $(3,4)$

ANSWER 25

7. $(2,1) \cdot (-1,2)$

ANSWER 0

Medium · the standard question

7 PROBLEMS

1. Angle between $(3,4)$ and $(1,2)$

ANSWER $\approx 10.3^\circ$

2. Angle between $(1,0)$ and $(1,1)$

ANSWER 45°

3. Angle between $(1,0)$ and $(-1,1)$

ANSWER 135°

4. Are $(2,1)$ and $(-1,2)$ perpendicular?

ANSWER Yes

5. t with $(t,3) \perp (4,-2)$

ANSWER 1.5

6. Is the angle between $(2,3)$ and $(-4,1)$ acute?

ANSWER No — the product is -5

7. $A(1,1)$, $B(4,2)$, $C(2,8)$: the right angle is at

ANSWER B

Hard · stretch and reason

7 PROBLEMS

1. Angle between $(1,1)$ and $(1,\sqrt{3})$

ANSWER 15°

2. $|\vec{a}| = 3$, $|\vec{b}| = 4$, $\varphi = 60^\circ$: $\vec{a} \cdot \vec{b}$

ANSWER 6

3. Same: $|\vec{a} + \vec{b}|$

ANSWER $\sqrt{37}$

4. $A(0,0)$, $B(4,3)$, $C(-3,4)$: the angle at A

ANSWER 90°

5. For which t is the angle between $(1,t)$ and $(1,1)$ equal to 45° ?

ANSWER $t = 0$

6. Prove the cosine rule using $|\vec{a} - \vec{b}|^2$

ANSWER Expand the scalar square

7. A rhombus: prove its diagonals are perpendicular

ANSWER $(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = |\vec{a}|^2 - |\vec{b}|^2 = 0$

07 Homework

Homework — 5 tasks

State the sign of the scalar product before computing the angle, as a check.

- Find $(5, 2) \cdot (3, -4)$ and say whether the angle is acute or obtuse.
- Find the angle between $(1, 1)$ and $(0, 1)$.
- For which t are $(t, 2)$ and $(3, -6)$ perpendicular?
- Show that $A(0, 0)$, $B(4, 3)$, $C(-3, 4)$ form a right angle at A .

5. Prove that the diagonals of a rhombus are perpendicular using the scalar product.

Solving triangles

Four kinds of data, four routes — and a checklist that finishes any triangle.

LESSON 28–29

2 × 40 MIN

GEOMETRIYA 9, §20

EXTENSION BEYOND IGX

LESSON CLOCK

13'

23'

22'

18'

4'

The four cases	13 min
SSS and SAS	23 min
ASA and SSA	22 min
Checking	18 min
Homework	4 min

LEARNING OBJECTIVES

- Recognise the four standard cases and choose the right route for each.
- Find all remaining sides and angles from three given elements.
- Decide how many triangles exist in the ambiguous case.
- Check the answer against the triangle inequality and the largest-side rule.

TEXTBOOK

National	pp. 111–117
Cambridge	Core & Extended, pp. 310–330
Workbook	Exercise 15.6

01

Explanation

The four cases

CASE	GIVEN	FIRST STEP	SOLUTIONS
SSS	three sides	cosine rule for the largest angle	1
SAS	two sides and the included angle	cosine rule for the third side	1
ASA / AAS	two angles and a side	third angle, then the sine rule	1
SSA	two sides and a non-included angle	sine rule for the second angle	0, 1 or 2

AAA IS NOT A CASE

Three angles determine the shape but not the size — infinitely many similar triangles fit. A question giving only angles cannot ask for a side.

SSS and SAS

SSS. $a = 5, b = 6, c = 7$.

STEP	WORKING	RESULT
largest angle first	$\cos C = \frac{25 + 36 - 49}{60} = 0.2$	$C \approx 78.5^\circ$
sine rule	$\sin A = \frac{5 \sin 78.5^\circ}{7}$	$A \approx 44.4^\circ$
angle sum	$180 - 78.5 - 44.4$	$B \approx 57.1^\circ$

SAS. $a = 8$, $b = 5$, $C = 60^\circ$. The cosine rule gives $c^2 = 64 + 25 - 40 = 49$, so $c = 7$; then the sine rule or the cosine rule finishes the angles.

FIND THE LARGEST ANGLE FIRST IN THE SSS CASE

It is the only one that might be obtuse. Once it is known, every remaining angle is acute, so the sine rule cannot be ambiguous — the ambiguity has been removed before it can arise.

ASA and SSA

ASA. $A = 50^\circ$, $B = 60^\circ$, $c = 10$. Then $C = 70^\circ$ and the sine rule gives $a = \frac{10 \sin 50^\circ}{\sin 70^\circ} \approx 8.15$ and $b = \frac{10 \sin 60^\circ}{\sin 70^\circ} \approx 9.21$.

SSA. This is the case that needs care. With a , b and A :

CONDITION	NUMBER OF TRIANGLES
$a \geq b$	1
$a < b$	compute $\sin B = \frac{b \sin A}{a}$
$\sin B > 1$	0
$\sin B = 1$	1
$\sin B < 1$	2

THE LAST ROW ASSUMES A IS ACUTE

If A is obtuse there can be at most one triangle, because a triangle has at most one obtuse angle. Check A before counting.

Checking

Three checks catch almost every arithmetic slip.

CHECK	STATEMENT
angle sum	the three angles total 180°
largest side	the largest side faces the largest angle
triangle inequality	each side is less than the sum of the other two

A fourth, when the area is also wanted: compute it twice, by $\frac{1}{2}ab \sin C$ and by Heron's formula, and compare.

THIRTY SECONDS OF CHECKING IS WORTH FIVE MARKS

An answer in which the largest angle faces the shortest side is wrong, whatever the working shows. That single glance catches mismatched pairs, the commonest error in this whole chapter.

02

Worked examples

EXAMPLE 1 Solve the triangle with $a = 5$, $b = 6$, $c = 7$.

1

$$\cos C = \frac{25 + 36 - 49}{60} = 0.2 \Rightarrow C \approx 78.5^\circ$$

Largest angle first.

2

$$\sin A = \frac{5 \sin 78.5^\circ}{7} \approx 0.700 \Rightarrow A \approx 44.4^\circ$$

3

$$B \approx 180 - 78.5 - 44.4 = 57.1^\circ$$

4

Check: largest side 7 faces 78.5° ✓

ANSWER

$$A \approx 44.4^\circ, B \approx 57.1^\circ, C \approx 78.5^\circ$$

EXAMPLE 2 Solve the triangle with $a = 8$, $b = 5$, $C = 60^\circ$.

$$1 \quad c^2 = 64 + 25 - 2 \cdot 40 \cdot 0.5 = 49$$

$$2 \quad c = 7$$

$$3 \quad \cos A = \frac{25 + 49 - 64}{70} = \frac{10}{70} \approx 0.143 \Rightarrow A \approx 81.8^\circ$$

$$4 \quad B \approx 180 - 60 - 81.8 = 38.2^\circ$$

ANSWER

$$c = 7, A \approx 81.8^\circ, B \approx 38.2^\circ$$

EXAMPLE 3 How many triangles have $a = 6$, $b = 10$, $A = 30^\circ$?

$$1 \quad a < b, \text{ and } A \text{ is acute — the ambiguous case.}$$

$$2 \quad \sin B = \frac{10 \times 0.5}{6} \approx 0.833$$

$$3 \quad B \approx 56.4^\circ \text{ or } 123.6^\circ.$$

$$4 \quad \text{Both leave a positive third angle — two triangles.}$$

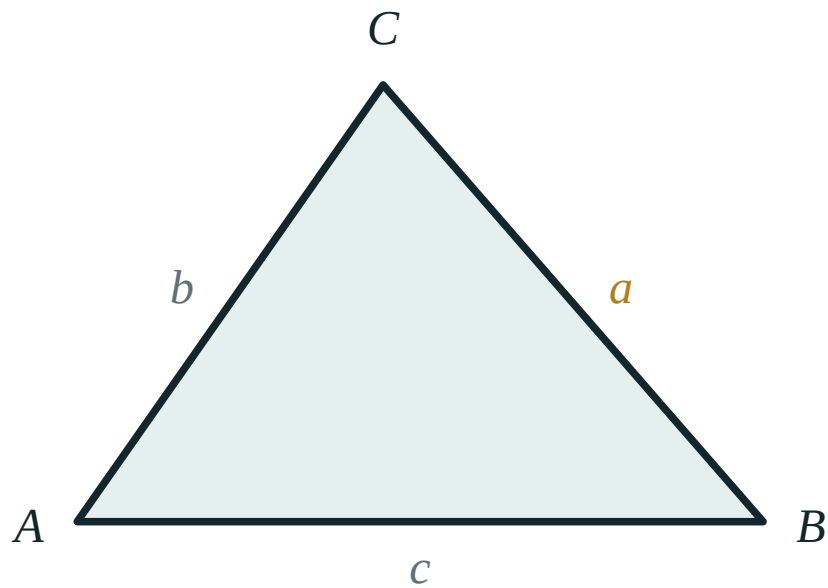
ANSWER

Two

03 Interactive model

Give the class one set of data at a time and ask only “which case, and which rule first?” for ten sets before any calculation; the recognition is the skill.

Three elements, one triangle



b 7

c 9

A 55°

a 7.6

B 49° C 76°

area 25.8

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
To solve a triangle	Uchburchakni yechish	Решить треугольник
Given elements	Berilgan elementlar	Данные элементы
Case	Hol	Случай
Triangle inequality	Uchburchak tengsizligi	Неравенство треугольника
Largest side	Eng katta tomon	Наибольшая сторона
Check	Tekshirish	Проверка
Two solutions	Ikkita yechim	Два решения
No solution	Yechimi yo'q	Нет решения

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

SSS starts with:

the sine rule

the cosine rule

the area formula

the angle sum

In SSS, find first:

the smallest angle

the largest angle

any angle

the area

ASA starts with:

the cosine rule

the third angle

the area

Pythagoras

SSA can give:

one triangle

two triangles

no triangle

any of these

AAA determines:

the triangle

the shape only

the area

nothing

The largest side faces:

the smallest angle

the largest angle

the right angle

no particular angle

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Three sides: which rule?

ANSWER Cosine

2. Two sides and the included angle: which rule?

ANSWER Cosine

3. Two angles and a side: which rule?

ANSWER Sine

4. Two sides and a non-included angle: which rule?

ANSWER Sine

5. $A = 50^\circ$, $B = 60^\circ$: C

ANSWER 70°

6. Largest side faces

ANSWER The largest angle

7. AAA determines

ANSWER The shape only

Medium · the standard question

7 PROBLEMS

1. $a = 5, b = 6, c = 7: C$

ANSWER $\approx 78.5^\circ$

2. Same: A

ANSWER $\approx 44.4^\circ$

3. $a = 8, b = 5, C = 60^\circ: c$

ANSWER 7

4. Same: A

ANSWER $\approx 81.8^\circ$

5. $A = 50^\circ, B = 60^\circ, c = 10: a$

ANSWER ≈ 8.15

6. Same: b

ANSWER ≈ 9.21

7. $a = 6, b = 10, A = 30^\circ$: how many triangles?

ANSWER Two

Hard · stretch and reason

7 PROBLEMS

1. $a = 5, b = 12, A = 60^\circ$: how many triangles?

ANSWER None

2. $a = 12, b = 5, A = 60^\circ$: how many triangles?

ANSWER One

3. $a = 5, b = 10, A = 30^\circ$: how many triangles?

ANSWER One — a right triangle

4. Solve $a = 7, b = 8, c = 9$: the largest angle

ANSWER $\approx 73.4^\circ$

5. Its area, by Heron

ANSWER ≈ 26.8

6. Its area, by $\frac{1}{2}ab \sin C$

ANSWER ≈ 26.8

7. A triangle with $A = 30^\circ, a = 6, b = 8$: the two possible areas

ANSWER ≈ 22.8 and ≈ 4.9

07 Homework

Homework — 5 tasks

Name the case before starting, and finish with the two checks.

1. Solve the triangle with $a = 6, b = 7, c = 9$.
2. Solve the triangle with $b = 10, c = 12, A = 45^\circ$.
3. Solve the triangle with $A = 40^\circ, C = 75^\circ, b = 15$.
4. How many triangles have $a = 7, b = 9, A = 50^\circ$?
5. Explain why the largest angle should be found first in the SSS case.

Chapter exercises

— Cambridge triangles and bearings

The same two rules, in the wording and the marking scheme of an IGCSE paper.

LESSON 30

1 × 40 MIN

GEOMETRIYA 9, II BOB MASHQLARI

IGX 3.2

LESSON CLOCK

8'

14'

12'

6'

What the wording asks for	8 min
Bearings both ways	14 min
Worked items	12 min
Homework	6 min

LEARNING OBJECTIVES

- Answer “calculate the length of...” and “calculate the size of...” in the expected form.
- Use three-figure bearings correctly in both directions.
- Give answers to 3 significant figures with units.
- Recognise when a diagram must be drawn before anything else.

TEXTBOOK

National	pp. 118–120
Cambridge	Core & Extended, pp. 51–58
Workbook	Exercise 3.2

01

Explanation

What the wording asks for

CAMBRIDGE SAYS	YOU MUST GIVE
“Calculate the length of BC ”	a number with units, 3 s.f.
“Calculate the size of angle ABC ”	degrees, 1 d.p. or 3 s.f.
“Show that $BC = 12.4$ ”	working ending at more accuracy than 12.4
“The diagram is not to scale”	never measure — calculate
“Calculate the bearing of C from A ”	three figures, e.g. 072°

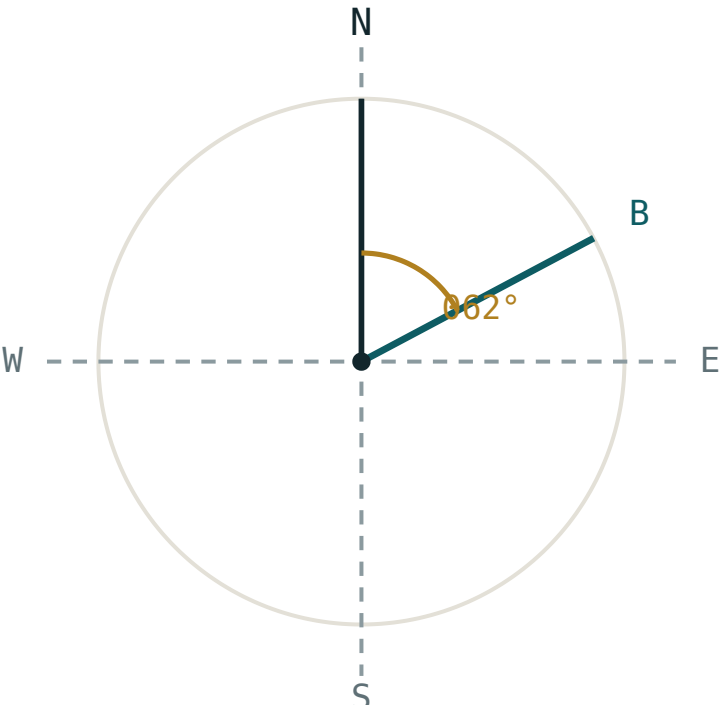
“SHOW THAT” MUST NOT START FROM THE ANSWER

Working that begins $BC = 12.4$ and verifies it scores nothing. Start from the data, and finish with a value more accurate than the one given — $12.43\dots$ — then state that it rounds to 12.4 .

Bearings both ways

Half the marks in a bearings question are for converting between the bearing and the angle inside the triangle.

SITUATION	ANGLE IN THE TRIANGLE
two bearings from one point	their difference
a leg and then a turn	$180^\circ - (\text{difference of the bearings})$
bearing of B from A is θ	bearing of A from B is $\theta \pm 180^\circ$



North is redrawn at every point; the angles of the triangle are read between the legs.

DRAW A NORTH ARROW AT EVERY VERTEX

Bearings are measured from north **at the point you are standing on**. Drawing the arrow at each vertex turns every bearing question into an ordinary angle-chase.

Worked items

ITEM	ANSWER
$AB = 9, AC = 12, \angle A = 65^\circ$: calculate BC	≈ 11.6
$a = 11, b = 13, c = 20$: calculate the largest angle	$\approx 113^\circ$
$A = 42^\circ, B = 71^\circ, a = 8$: calculate b	≈ 11.3
$AB = 9, AC = 12, \angle A = 65^\circ$: calculate the area	≈ 48.9

Each answer is given to three significant figures, with the units the question used.

02 Worked examples

EXAMPLE 1 In $\triangle ABC$, $AB = 9\text{ cm}$, $AC = 12\text{ cm}$, $\angle A = 65^\circ$. Calculate BC .

- 1 SAS — use the cosine rule.
- 2 $BC^2 = 81 + 144 - 2 \cdot 108 \cdot \cos 65^\circ$
- 3 $= 225 - 216 \times 0.4226 = 133.7$
- 4 $BC \approx 11.6\text{ cm}$
Three significant figures.

ANSWER

$\approx 11.6\text{ cm}$

EXAMPLE 2 A ship sails from A to B on a bearing of 072° . What is the bearing of A from B ?

- ① Add 180° .
- ② $072^\circ + 180^\circ = 252^\circ$
- ③ Less than 360° , so no subtraction is needed.
- ④ 252°
Three figures.

ANSWER

252°

EXAMPLE 3 Calculate the area of the triangle with $AB = 9$, $AC = 12$ and $\angle A = 65^\circ$.

- ① $S = \frac{1}{2} \cdot 9 \cdot 12 \cdot \sin 65^\circ$
- ② $= 54 \times 0.9063$
- ③ ≈ 48.9
- ④ Units: cm^2 if the sides were in cm .

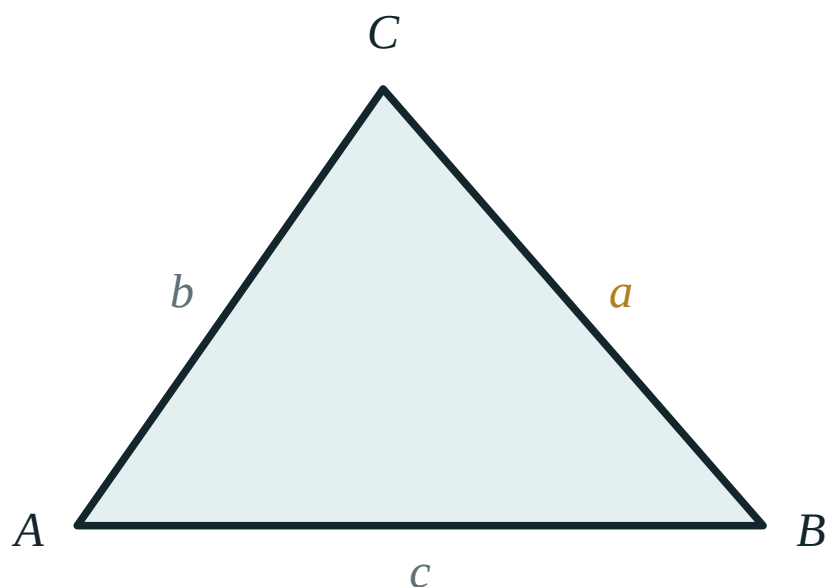
ANSWER

$\approx 48.9 \text{ cm}^2$

03 Interactive model

Show a “show that” answer that begins with the given value and one that begins with the data; the class marks both, and the rule becomes theirs.

Calculate and round



b 7

c 9

A 55°

a 7.6

B 49°

C 76°

area 25.8

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Calculate	Hisoblang	Вычислите
Three significant figures	Uchta ma’noli raqam	Три значащие цифры
Not to scale	Masshtabda emas	Не в масштабе
Bearing	Azimut	Азимут
Correct to	Aniqlikda	С точностью до
Show your working	Yechimni yozing	Покажите решение
Units	Birliklar	Единицы
Diagram	Chizma	Диаграмма

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Answers are normally given to:

1 s.f.

2 s.f.

3 s.f.

exact values only

“Not to scale” means:

measure carefully

calculate

estimate

redraw

A “show that” answer must:

start from the answer

start from the data

be measured

be estimated

Bearing of B from A is 072° . Bearing of A from B :

$AB = 9$, $AC = 12$, $\angle A = 65^\circ$: $BC \approx$

A bearing is always written with:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Bearing of due west

ANSWER 270°

2. Back bearing of 072°

ANSWER 252°

3. Back bearing of 250°

ANSWER 070°

4. Answers to how many significant figures?

ANSWER Three

5. “Not to scale” means

ANSWER Calculate, do not measure

6. $AB = 9$, $AC = 12$, $\angle A = 65^\circ$: which rule?

ANSWER Cosine

7. $A = 42^\circ$, $B = 71^\circ$, $a = 8$: which rule?

ANSWER Sine

Medium · the standard question

7 PROBLEMS

1. $AB = 9, AC = 12, \angle A = 65^\circ$: BC

ANSWER ≈ 11.6

2. Same: the area

ANSWER ≈ 48.9

3. $a = 11, b = 13, c = 20$: the largest angle

ANSWER $\approx 113^\circ$

4. $A = 42^\circ, B = 71^\circ, a = 8$: b

ANSWER ≈ 11.3

5. $a = 7, b = 10, C = 55^\circ$: c

ANSWER ≈ 8.29

6. $a = 6, b = 7, c = 8$: the smallest angle

ANSWER $\approx 46.6^\circ$

7. A ship on 072° for 40 km : how far east?

ANSWER $\approx 38.0\text{ km}$

Hard · stretch and reason

7 PROBLEMS

1. B is 12 km from A on 040° ; C is 15 km from B on 130° : AC

ANSWER ≈ 19.2 km

2. Its bearing from A

ANSWER $\approx 091^\circ$

3. A triangle with $a = 9$, $b = 12$, $c = 15$: its type

ANSWER Right-angled

4. Its area

ANSWER 54

5. Show that $BC = 11.6$ when $AB = 9$, $AC = 12$, $\angle A = 65^\circ$

ANSWER $BC = 11.56\dots$ rounds to 11.6

6. A kite: two pairs 6, 6 and 9, 9 with an angle of 120° between 6 and 9

ANSWER Area ≈ 46.8

7. A regular pentagon of side 10: its area

ANSWER ≈ 172

07 Homework

Homework — 5 tasks

Every answer needs units and three significant figures unless the question says otherwise.

- In $\triangle ABC$, $AB = 7$ cm, $AC = 10$ cm, $\angle A = 48^\circ$. Calculate BC and the area.
- Calculate the largest angle of the triangle with sides 9, 12 and 18 cm.
- A ship sails from P on a bearing of 115° . Find the bearing of P from the ship.
- In $\triangle ABC$, $A = 38^\circ$, $B = 64^\circ$, $a = 11$ cm. Calculate b .
- Show that a triangle with sides 9, 12 and 15 is right-angled, and find its area.

Control work 2, and work on the mistakes

Solving triangles in one paper, and the trigonometry chapter closed.

LESSON 31–32

2 × 40 MIN

GEOMETRIYA 9, NAZORAT ISHI 2

IGX 3, 7 REVIEW

LESSON CLOCK

3

40'

12'

20'

5'

Instructions

3 min

The paper

40 min

Answers

12 min

Diagnosis and rewrite

20 min

The map

5 min

LEARNING OBJECTIVES

- Choose and apply the correct rule under time.
- Use the area formula and the extended ratios accurately.
- Handle the ambiguous case correctly.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National

pp. 76–120

Cambridge

Core & Extended, pp. 40–72, 130–140

Workbook

Control paper G2

01

Explanation

The paper — 30 marks, 40 minutes

Q	TASK	MARKS	FROM
1	Find $\sin 135^\circ$, $\cos 135^\circ$ and $\tan 150^\circ$	5	L19–20
2	Find the area of a triangle with sides 9 and 12 and included angle 30°	4	L21
3	$A = 45^\circ$, $B = 60^\circ$, $a = 10$: find b	5	L22
4	$a = 6$, $b = 9$, $C = 120^\circ$: find c	5	L23
5	Find the largest angle of the triangle with sides 4, 7, 9	5	L23
6	Find the angle between $(2, 1)$ and $(1, 3)$	6	L26–27

WHERE THE MARKS ACTUALLY GO

Q1 carries two marks for the signs; Q3 one for finding C first; Q4 one for the sign of $\cos 120^\circ$; Q5 two for choosing the angle opposite 9; Q6 two for the two magnitudes.

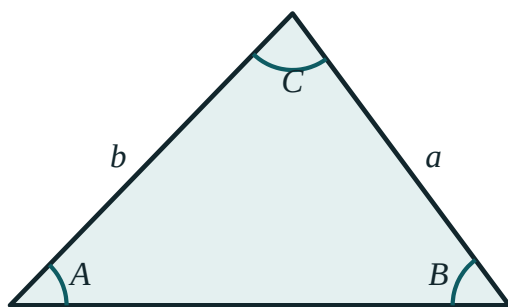
Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
cosine of an obtuse angle left positive	$\cos 135^\circ = \frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2}$
angle not the included one	$\frac{1}{2}ab \sin A$	$\frac{1}{2}ab \sin C$
mismatched pair in the sine rule	$\frac{a}{\sin B}$	$\frac{a}{\sin A}$
sign of the correction term	$c^2 = 36 + 81 - 54$	$= 36 + 81 + 54$
wrong angle chosen as largest	opposite 4	opposite 9
magnitudes omitted	$\cos \varphi = \vec{a} \cdot \vec{b}$	$\cos \varphi = \frac{\vec{a} \cdot \vec{b}}{ \vec{a} \vec{b} }$
second answer missing in SSA	one angle	test the obtuse option

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

Chapter II as one map

BLOCK	THE SENTENCE
ratios to 180°	coordinates on the upper semicircle; sine positive, cosine changing sign
the area	$S = \frac{1}{2}ab \sin C$
the sine rule	$\frac{a}{\sin A} = 2R$ — <i>needs a matched pair</i>
the cosine rule	Pythagoras with $-2ab \cos C$
solving a triangle	four cases, and only SSA is ambiguous
the scalar product	$\vec{a} \cdot \vec{b} = x_1x_2 + y_1y_2 = \vec{a} \vec{b} \cos \varphi$



$$a / \sin A = b / \sin B = c / \sin C$$

$$a^2 = b^2 + c^2 - 2bc \cdot \cos A$$

One triangle, two rules — and between them every triangle can be solved.

LOOKING FORWARD

Chapter III returns to the circle: inscribed and circumscribed polygons, regular polygons, and the circumference and area of a disc. The trigonometry of this chapter is used throughout — the side of a regular n -gon is found with a sine.

02 Worked examples

EXAMPLE 1 Model answer, Q4: $a = 6$, $b = 9$, $C = 120^\circ$.

$$1 \quad c^2 = 36 + 81 - 2 \cdot 54 \cdot \cos 120^\circ$$

$$2 \quad \cos 120^\circ = -0.5$$

The sign matters.

$$3 \quad c^2 = 117 + 54 = 171$$

$$4 \quad c = \sqrt{171} \approx 13.1$$

ANSWER

$$c \approx 13.1$$

EXAMPLE 2 Model answer, Q5: the largest angle of 4, 7, 9.

① It is opposite the longest side, 9.

② $\cos C = \frac{16 + 49 - 81}{2 \cdot 4 \cdot 7}$

③ $= \frac{-16}{56} \approx -0.2857$

Negative — obtuse.

④ $C \approx 106.6^\circ$

ANSWER

$\approx 106.6^\circ$

EXAMPLE 3 Model answer, Q6: the angle between (2, 1) and (1, 3).

① $\vec{a} \cdot \vec{b} = 2 + 3 = 5$

② $|\vec{a}| = \sqrt{5}, |\vec{b}| = \sqrt{10}$

③ $\cos \varphi = \frac{5}{\sqrt{50}} \approx 0.7071$

④ $\varphi = 45^\circ$

Exactly.

ANSWER

45°

03 Interactive model

Read out a solution to Q4 that treats $\cos 120^\circ$ as positive; the class finds a c smaller than both given sides, which is impossible for an obtuse angle.

Chapter II in twelve questions

$\sin(180^\circ - \alpha)$ equals:

$\cos 120^\circ$ equals:

The area formula is:

The area is greatest when C is:

30°

60°

90°

120°

The sine rule needs:

three sides

a matched pair

a right angle

an area

The common value of the sine rule is:

R

$2R$

πR

R^2

The cosine rule is used for:

AAS

SSA

SAS or SSS

AAA

$\cos C < 0$ means:

acute

right

obtuse

impossible

Only which case is ambiguous?

SSS

SAS

ASA

SSA

The largest side faces:

the smallest angle

the largest angle

the right angle

no particular angle

$\vec{a} \cdot \vec{b}$ in coordinates:

$$x_1 y_2 - x_2 y_1$$

$$x_1 x_2 + y_1 y_2$$

$$x_1 x_2 - y_1 y_2$$

$$x_1 + y_1$$

$\vec{a} \cdot \vec{b} = 0$ means:

parallel

perpendicular

equal

zero

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Sine rule	Sinuslar teoremasi	Теорема синусов
Cosine rule	Kosinuslar teoremasi	Теорема косинусов
Area formula	Yuza formulasi	Формула площади
Scalar product	Skalyar ko'paytma	Скалярное произведение
Ambiguous case	Noaniq hol	Неоднозначный случай
Bearing	Azimut	Азимут
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 loses marks most often for:

arithmetic

the signs

the formula

the units

Q2's angle must be:

the largest

the included one

acute

opposite a

Q3 begins by finding:

b

C

the area

R

In Q4 the correction term is:

subtracted

added

zero

halved

Q5's angle is opposite:

4

7

9

any side

Q6 needs, besides the product:

an angle

the two magnitudes

a diagram

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\sin 135^\circ$

ANSWER $\frac{\sqrt{2}}{2}$

2. $\cos 135^\circ$

ANSWER $-\frac{\sqrt{2}}{2}$

3. $\tan 150^\circ$

ANSWER $-\frac{\sqrt{3}}{3}$

4. Sides 9, 12, angle 30° : area

ANSWER 27

5. $A = 45^\circ, B = 60^\circ$: C

ANSWER 75°

6. $(2,1) \cdot (1,3)$

ANSWER 5

7. $|(2,1)|$

ANSWER $\sqrt{5}$

Medium · the standard question

7 PROBLEMS

1. $A = 45^\circ, B = 60^\circ, a = 10: b$

ANSWER ≈ 12.2

2. $a = 6, b = 9, C = 120^\circ: c$

ANSWER ≈ 13.1

3. Sides 4, 7, 9: the largest angle

ANSWER $\approx 106.6^\circ$

4. Angle between (2,1) and (1,3)

ANSWER 45°

5. Sides 4, 7, 9: the area

ANSWER ≈ 13.4

6. $a = 10, A = 45^\circ: R$

ANSWER ≈ 7.07

7. Sides 4, 7, 9: type

ANSWER Obtuse

Hard · stretch and reason

7 PROBLEMS

1. A triangle with $a = 8$, $b = 11$, $A = 40^\circ$: how many triangles?

ANSWER Two

2. Its two possible areas

ANSWER ≈ 43.0 and ≈ 16.5

3. A parallelogram 6, 10 with an angle of 135° : the shorter diagonal

ANSWER ≈ 7.15

4. For which t are $(t, 4)$ and $(2, -t)$ perpendicular?

ANSWER $t = 0$

5. A triangle inscribed in a circle of radius 8 with $A = 60^\circ$: a

ANSWER $8\sqrt{3}$

6. A ship: 18 km on 050° then 24 km on 140°

ANSWER 30 km

7. Sides 5, 5, 6: the area

ANSWER 12

07 Homework

Homework — 5 tasks

Rewrite in full every question that lost a mark before Chapter III begins.

- Find $\sin 120^\circ$, $\cos 150^\circ$ and $\tan 135^\circ$.
- Find the area of a triangle with sides 8 and 15 and included angle 150° .
- In $\triangle ABC$, $A = 35^\circ$, $B = 80^\circ$, $a = 9$. Find b .
- Find the largest angle of the triangle with sides 5, 9, 11.
- Find the angle between the vectors $(3, 1)$ and $(1, 2)$.